

ModSwap — KMUTT Student Marketplace

Project Report

Concept · Persona · Journey · WBS · Backlog · Gantt · Precedence · Architecture

Team

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Team Members

Name	Role
Sorasit Laiget	Backend Lead · Backend Integration
Veerachai Sribuatim	UX/UI Design · Frontend
Poowrin Trijakpranee	Frontend
Chakkrapat Supanith	UX/UI Design · Frontend · Backend
Siwakon Wanichsujit	Frontend

1. Concept and List of Functions

Concept

ModSwap is a KMUTT-exclusive student-to-student marketplace mobile application that enables students to buy, sell, and swap second-hand items within the KMUTT community. Access is restricted to verified `@mail.kmutt.ac.th` email addresses, ensuring a safe and trusted campus environment.

The app addresses the common problem of students accumulating unused textbooks, electronics, and equipment between academic terms, while other students need the same items at lower prices. ModSwap creates a closed-loop circular economy within the campus.

Key Value Propositions

- **Trust** — Only verified KMUTT students can participate
- **Convenience** — AI-powered semantic search finds items intelligently
- **Safety** — Seller ratings and deal history build community trust
- **Flexibility** — Supports sale, swap, or both listing types

List of Functions

Authentication & Onboarding

#	Function	Description
A1	Register	Create account with KMUTT email, verify email
A2	Login	Email/password sign-in with Firebase Auth
A3	Biometric Login	Face ID / Touch ID via <code>local_auth</code>
A4	Complete Profile	Set display name, student ID, LINE ID after first login
A5	Email Verification	Only <code>@mail.kmutt.ac.th</code> emails allowed; verified via Firebase
A6	Change Password	Secure password update from settings
A7	Delete Account	Soft-delete flow; removes user data

Listings

#	Function	Description
L1	Browse Listings	Home feed with category filter tabs
L2	Semantic Search	AI-powered search using Gemini embedding-001 (768-dim cosine similarity)
L3	Keyword Search	Fallback text search when semantic search is disabled
L4	Category Filter	Filter by Books, Electronics, Clothes, Stationery, etc.
L5	Listing Detail	Full item view: images, description, price, seller info, condition
L6	Post Item	Create listing with images, price, swap preferences, condition
L7	Edit Listing	Modify unpublished/published items

#	Function	Description
L8	Delete Listing	Remove item from marketplace
L9	My Items	View own listings by state: Draft / Published / Sold
L10	Image Upload	Multi-image upload with compression to Firebase Storage
L11	Mark as Sold	Record completed deal with buyer LINE ID, price, deal type
L12	View Count	Track how many times each listing was viewed

Social & Trust

#	Function	Description
S1	Wishlist	Save/unsave listings; view saved items
S2	Seller Profile	View any seller's profile, rating, listings
S3	Rate Seller	Submit 1–5 star rating with review text after deal
S4	View Ratings	See all ratings received on own profile
S5	Pending Rating	System prompts buyer to rate seller after deal completes

Notifications

#	Function	Description
N1	Wishlist Notification	Alert when someone wishlists your item
N2	Rating Received	Alert when you receive a new rating
N3	Rate Request	Prompt buyer to rate after deal is marked complete
N4	Mark Sold Reminder	Confirmation when deal is recorded
N5	Price Drop	Alert when a wishlisted item's price drops
N6	Trending	Recommendation for popular items in preferred categories
N7	Welcome	Sent on first login
N8	Email Verified	Confirmation of email verification
N9	Security Alert	New device login detection
N10	Password Changed	Confirmation of password update

System / Settings

#	Function	Description
P1	Edit Profile	Update display name, student ID, LINE ID, avatar
P2	Theme Toggle	Light / Dark mode
P3	Feature Flags	Firebase Remote Config controls semantic search toggle
P4	Logout	Sign out and clear session

2. User Persona

Primary Persona

Name: Nattawut Charoensuk **Nickname:** Net **Age:** 20 years old **Year:** 2nd Year **Faculty:** Faculty of Engineering, KMUTT **Major:** Computer Engineering

Background: Net is a sophomore at KMUTT who lives in a student dormitory near the Bang Mod campus. He receives a monthly allowance of 6,000 THB from his parents and takes on occasional freelance web development work. He is tech-savvy, uses his phone for most transactions, and is comfortable with apps like Shopee, LINE, and Facebook Marketplace.

Goals: - Find affordable second-hand engineering textbooks at the start of each semester - Sell textbooks and equipment he no longer uses to recoup costs - Swap his old iPad for a newer model before his 3rd year

Frustrations: - Facebook Marketplace groups are full of non-KMUTT sellers; hard to coordinate meetups - No way to verify if a seller is a real KMUTT student - Manually browsing hundreds of listings is time-consuming - Sellers often ghost buyers or don't reply promptly

Tech Behavior: - Uses phone 6–8 hours/day - Prefers chat (LINE) over phone calls - Pays with PromptPay / QR code - Takes photos of items before buying

Quote: > "I want to buy second-hand items at uni safely, without worrying about being scammed — meet up right here on campus."

Device: iPhone 14 (iOS 17), MacBook Air M2 **Preferred Payment:** PromptPay, Cash **Meetup**

Preference: On-campus (Engineering canteen, Library, Dorm lobby)

Secondary Persona

Name: Wanwisa Thongprasert **Nickname:** Fah **Age:** 22 years old **Year:** 4th Year **Faculty:** Faculty of Architecture and Design

Background: Fah is graduating and needs to clear out 4 years' worth of architecture supplies, drawing tools, and textbooks. She wants quick sales before she moves out of her dormitory.

Goals: - Sell items quickly at fair prices - Build a seller reputation so buyers trust her listings - Minimal friction — post item in under 5 minutes

Frustrations: - No centralized KMUTT marketplace before ModSwap - Buyers don't show up after agreeing to meet

3. User Journey Map

Persona: Nattawut (Net), 2nd-year Engineering student **Goal:** Find and purchase a second-hand Calculus textbook

Phase	Step	User Action	System Response	Emotion	Pain Point
1. Discover	1.1	Hears about ModSwap from a friend	—	Curious	—
	1.2	Downloads app from App Store	App opens to Login screen	Hopeful	Slow download on campus WiFi
2. Register	2.1	Taps “Register”	Registration form shown	Neutral	—
	2.2	Enters KMUTT email + password	Form validates email domain	Confident	Error if non-KMUTT email
	2.3	Checks email for verification link	Firebase sends verification email	Waiting	Email takes 1–2 min
	2.4	Taps verification link	Account verified; redirect to Complete Profile	Relieved	—
	2.5	Fills in display name, student ID, LINE ID	Profile saved; navigates to Home	Excited	—
3. Browse & Search	3.1	Sees home feed	Listings loaded (published state)	Engaged	Too many items, hard to find
	3.2	Types “Calculus” in search bar	Semantic search returns relevant results	Impressed	—
	3.3	Taps category “Books”	Filtered feed shows only book listings	Satisfied	—
	3.4	Taps listing “Calculus: Early Transcendentals”	Listing detail screen opens	Interested	—
4. Contact & Negotiate	4.1	Views listing: images, price (250 THB), seller rating	Full details shown	Positive	Wishes there were more photos
	4.2	Taps “Wishlist” heart icon	Listing saved; wishlist count increments	Committed	—
	4.3	Notes seller’s LINE ID; contacts via LINE	— (outside app)	Slightly frustrated	Has to leave app to contact
	4.4	Agrees to meet at Engineering canteen	—	Excited	—
5. Complete Deal	5.1	Meets seller on campus; inspects item	—	Satisfied	—

Phase	Step	User Action	System Response	Emotion	Pain Point
	5.2	Pays 230 THB (negotiated) via PromptPay	—	Happy	—
	5.3	Seller marks listing as sold in app	Listing state → 'sold'; Deal record created	—	—
	5.4	Net receives "Deal Complete!" notification	<code>markSoldReminder</code> notification in app	Notified	—
6. Rate & Return	6.1	Receives "Rate Your Experience" prompt	<code>rateRequest</code> notification shown	Reminded	—
	6.2	Opens notification → Rating screen	Rating form loaded	Engaged	—
	6.3	Submits 5-star rating with comment	Rating saved; seller's avg rating updated	Accomplished	—
	6.4	Returns to browse more listings	Home feed	Loyal	—

Moments of Delight: - Semantic search correctly finds Thai-language items - On-campus meetup reduces risk of being scammed - Seller rating system builds trust before committing

Moments of Friction: - Contact outside app (LINE) breaks the in-app flow - Verification email delay during registration

4. Work Breakdown Structure (WBS)

ID	Task Description	Est.	Assignee
1. Project Management			
1.1	Requirements Gathering & Analysis	8h	Sorasit · Veerachai
1.2	Sprint Planning & Backlog Grooming	4h	Sorasit
1.3	Weekly Stand-ups & Progress Tracking	6h	All
1.4	Final Report & Documentation	8h	All
2. UI/UX Design			
2.1	User Research & Persona Creation	4h	Veerachai · Chakkrapat
2.2	User Journey Mapping	3h	Veerachai
2.3	Wireframes (Lo-Fi)	6h	Veerachai · Chakkrapat
2.4	High-Fidelity Mockups (Figma)	10h	Veerachai · Chakkrapat
2.5	Design System & Component Library	4h	Veerachai
3. Authentication & User Management			
3.1	Firebase Auth Setup	2h	Sorasit
3.2	Register / Login / Logout Flow	6h	Siwakon
3.3	Email Verification (KMUTT domain guard)	3h	Sorasit
3.4	Complete Profile Screen	4h	Siwakon
3.5	Biometric Authentication	5h	Siwakon
3.6	Change Password / Security Settings	3h	Siwakon
4. Listings Module			
4.1	Firestore Schema Design	3h	Sorasit
4.2	Post Item Screen (Create/Edit)	10h	Poowrin
4.3	Image Upload with Compression	4h	Poowrin
4.4	Home Feed (Browse + Category Filter)	8h	Poowrin
4.5	Listing Detail Screen	8h	Chakkrapat
4.6	My Items Screen (Draft/Published/Sold)	6h	Chakkrapat
4.7	Mark as Sold Flow	6h	Chakkrapat · Sorasit
5. Search			
5.1	Gemini Embedding Integration	6h	Sorasit
5.2	Cloud Function: Generate Embedding	4h	Sorasit
5.3	Cosine Similarity Search Endpoint	6h	Sorasit
5.4	Search UI (Search Bar + Results)	5h	Poowrin
5.5	Remote Config Feature Flag	2h	Sorasit
6. Social & Trust Features			

ID	Task Description	Est.	Assignee
6.1	Wishlist (Add/Remove + Count Badge)	4h	Siwakon
6.2	Seller Profile View	5h	Chakkrapat
6.3	Rating System (Submit + Display)	8h	Poowrin · Sorasit
6.4	Pending Rating Flow	5h	Sorasit
6.5	Edit Profile	4h	Siwakon
7. Notifications			
7.1	Firestore Notification Schema	2h	Sorasit
7.2	Notification Model & Types	3h	Chakkrapat
7.3	Notification Screen (Tabs + Grouping)	6h	Chakkrapat
7.4	NotificationService (Read/Stream)	4h	Sorasit
7.5	Unread Badge on Nav Bar	2h	Chakkrapat
8. Backend (Cloud Functions)			
8.1	Express API Setup + Middleware	4h	Sorasit
8.2	Auth Middleware (Firebase Token Verify)	2h	Sorasit
8.3	Listings CRUD Endpoints	8h	Sorasit
8.4	Deals Service (Mark as Sold + Stats)	6h	Sorasit
8.5	Ratings Service	4h	Sorasit
8.6	Firestore Triggers (onCreate/onDelete)	4h	Sorasit
8.7	Firestore Security Rules	4h	Sorasit
9. QA, Testing & Deployment			
9.1	Unit Tests (Services & Utilities)	6h	Sorasit
9.2	Widget Tests	6h	Poowrin
9.3	Integration Tests	4h	Poowrin
9.4	Manual QA Testing	4h	All
9.5	Firebase Emulator Testing	3h	Sorasit
9.6	Build & Deployment	3h	Sorasit

Total Estimated Effort: ~220 person-hours

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5. Backlog Planning

Sprint 1 — Foundation (Week 1)

Goal: Project setup, authentication, core infrastructure

ID	User Story	Priority	Points	Assignee
US-01	As a student, I want to register with my KMUTT email so that only real KMUTT students can access the app	Must	5	Siwakon
US-02	As a user, I want to verify my KMUTT email before I can use the app	Must	3	Sorasit
US-03	As a user, I want to complete my profile (name, student ID, LINE ID) after registration	Must	3	Siwakon
US-04	As a user, I want to log in and log out securely	Must	2	Siwakon
US-05	As a developer, I want Firebase Auth + Cloud Functions backend with auth middleware	Must	5	Sorasit
US-06	As a developer, I want Firestore schema designed for users, listings, deals, notifications	Must	3	Sorasit

Sprint 1 Velocity: 21 points

Sprint 2 — Core Listings (Week 2)

Goal: Browse, post, and manage listings

ID	User Story	Priority	Points	Assignee
US-07	As a seller, I want to post a listing with images, price, and description	Must	8	Powwrin
US-08	As a buyer, I want to browse listings on the home feed	Must	5	Powwrin
US-09	As a buyer, I want to filter listings by category	Must	3	Powwrin
US-10	As a buyer, I want to view the full listing detail with all images	Must	5	Chakkrapat

ID	User Story	Priority	Points	Assignee
US-11	As a seller, I want to edit and delete my listings	Should	3	Chakkrapat
US-12	As a seller, I want to manage my listings by state (draft/published/sold)	Should	3	Chakkrapat

Sprint 2 Velocity: 27 points

Sprint 3 — Search & Social (Week 3)

Goal: AI search, wishlist, seller profiles, ratings

ID	User Story	Priority	Points	Assignee
US-13	As a buyer, I want to search for items using natural language (AI semantic search)	Must	8	Sorasit
US-14	As a buyer, I want to save listings to my wishlist	Should	3	Siwakon
US-15	As a buyer, I want to view a seller's profile and ratings before buying	Must	5	Chakkrapat
US-16	As a buyer, I want to rate a seller after a deal completes	Must	5	Poowrin
US-17	As a seller, I want to mark a deal as sold and record buyer details	Must	5	Chakkrapat
US-18	As a developer, I want Firebase Remote Config to toggle semantic search on/off	Should	2	Sorasit

Sprint 3 Velocity: 28 points

Sprint 4 — Notifications & Security (Week 4)

Goal: Full notification system, biometric auth, security

ID	User Story	Priority	Points	Assignee
US-19	As a user, I want in-app notifications grouped by category (Buyer / For You / System)	Must	8	Chakkrapat

ID	User Story	Priority	Points	Assignee
US-20	As a seller, I want to be notified when someone wishlists my item	Should	3	Sorasit
US-21	As a buyer, I want to be prompted to rate a seller after a deal	Must	3	Sorasit
US-22	As a user, I want to log in with Face ID / Touch ID for convenience	Should	5	Siwakon
US-23	As a user, I want to change my password securely	Should	2	Siwakon
US-24	As a user, I want to update my profile (name, avatar, LINE ID)	Should	3	Siwakon

Sprint 4 Velocity: 24 points

Sprint 5 — Polish, Testing & Deployment (Week 5)

Goal: QA, bug fixes, performance, documentation

ID	User Story	Priority	Points	Assignee
US-25	As a developer, I want widget tests for core screens	Must	5	Powwrin
US-26	As a developer, I want integration tests for auth and listing flows	Should	5	Powwrin
US-27	As a user, I want dark/light theme toggle	Should	2	Veerachai
US-28	As a developer, I want Firestore security rules that restrict access properly	Must	3	Sorasit
US-29	As a team, I want all project documentation completed	Must	5	All
US-30	As a developer, I want the app builds and runs on iOS and Android	Must	3	Sorasit

Sprint 5 Velocity: 23 points

Total Backlog: 123 story points across 30 user stories

6. Gantt Chart

Task / Feature	Sprint 1 W1 Mar 1–7	Sprint 2 W2 Mar 8–14	Sprint 3 W3 Mar 15–21	Sprint 4 W4 Mar 22–28	Sprint 5 W5 Mar 29–Apr 4
Project Setup & UX Design Sorasit · Veerachai · Chakkrapat					
Authentication & Onboarding Siwakon · Sorasit					
Backend Infrastructure (API + Auth) Sorasit					
Listings Module (CRUD + Feed) Poowrin · Chakkrapat					
AI Semantic Search (Gemini) Sorasit · Poowrin					
Social Features (Wishlist + Ratings) Siwakon · Chakkrapat · Poowrin					
Mark as Sold & Deal Flow Chakkrapat · Sorasit					
Notification System Chakkrapat · Sorasit					
Biometric Auth & Security Siwakon · Sorasit					
Testing & QA Poowrin · Sorasit					
Documentation & Deployment All					

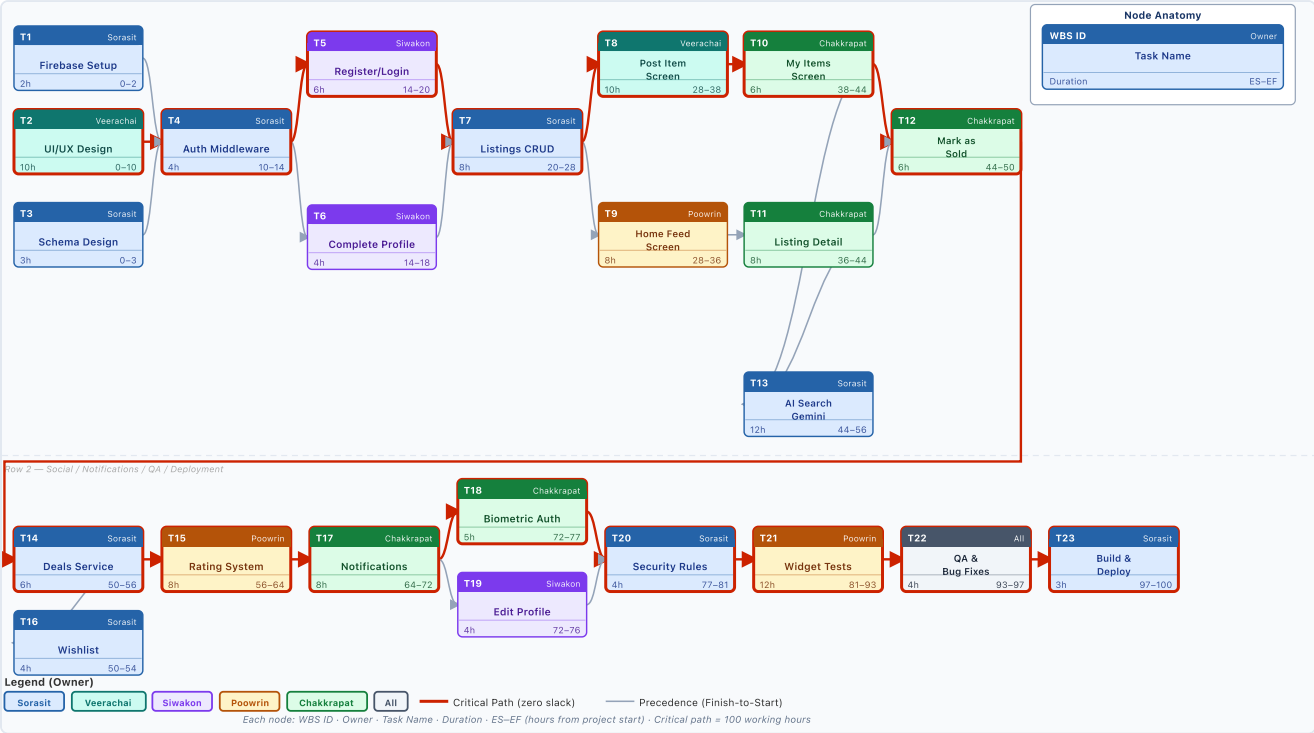
 Sprint 1 (W1 Mar 1–7)  Sprint 2 (W2 Mar 8–14)  Sprint 3 (W3 Mar 15–21)  Sprint 4 (W4 Mar 22–28)
 Sprint 5 (W5 Mar 29–Apr 4)

Legend:  = Active development period

Milestones: - **M1 (End W1):** Auth flow complete + Firebase connected - **M2 (End W2):** Core listing CRUD working - **M3 (End W3):** Search + Social features live - **M4 (End W4):** Full notifications + biometric auth - **M5 (End W5):** Tests, security rules, deployment ready

7. Precedence Diagram (Task Dependencies)

Precedence Diagram — ModSwap · 23 tasks · 5 sprints · Critical path highlighted in red (100 working hours)

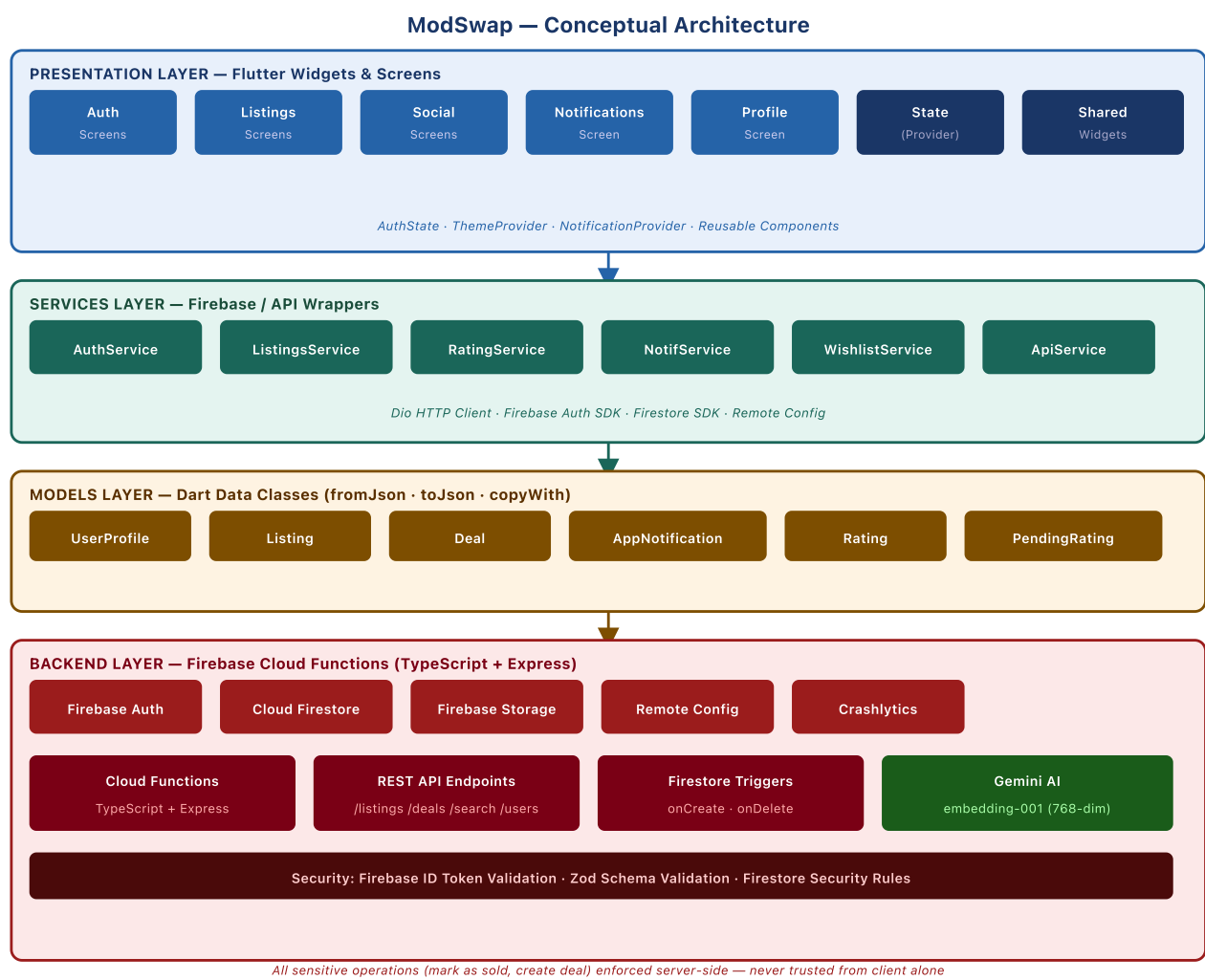


Critical Path: T1 → T4 → T5 → T7 → T8 → T10 → T12 → T14 → T15 → T17 → T18 → T20 → T21 → T22 → T23

Critical Path Duration: ~90 hours

Parallel Tracks: - T2 (UI/UX Design) runs parallel to T1, T3, T4 - T13 (AI Search) runs parallel to T12 (Mark as Sold) - T16 (Wishlist) runs parallel to T15 (Ratings) - T19 (Edit Profile) runs parallel to T18 (Biometric Auth)

8. Conceptual Architecture Diagram



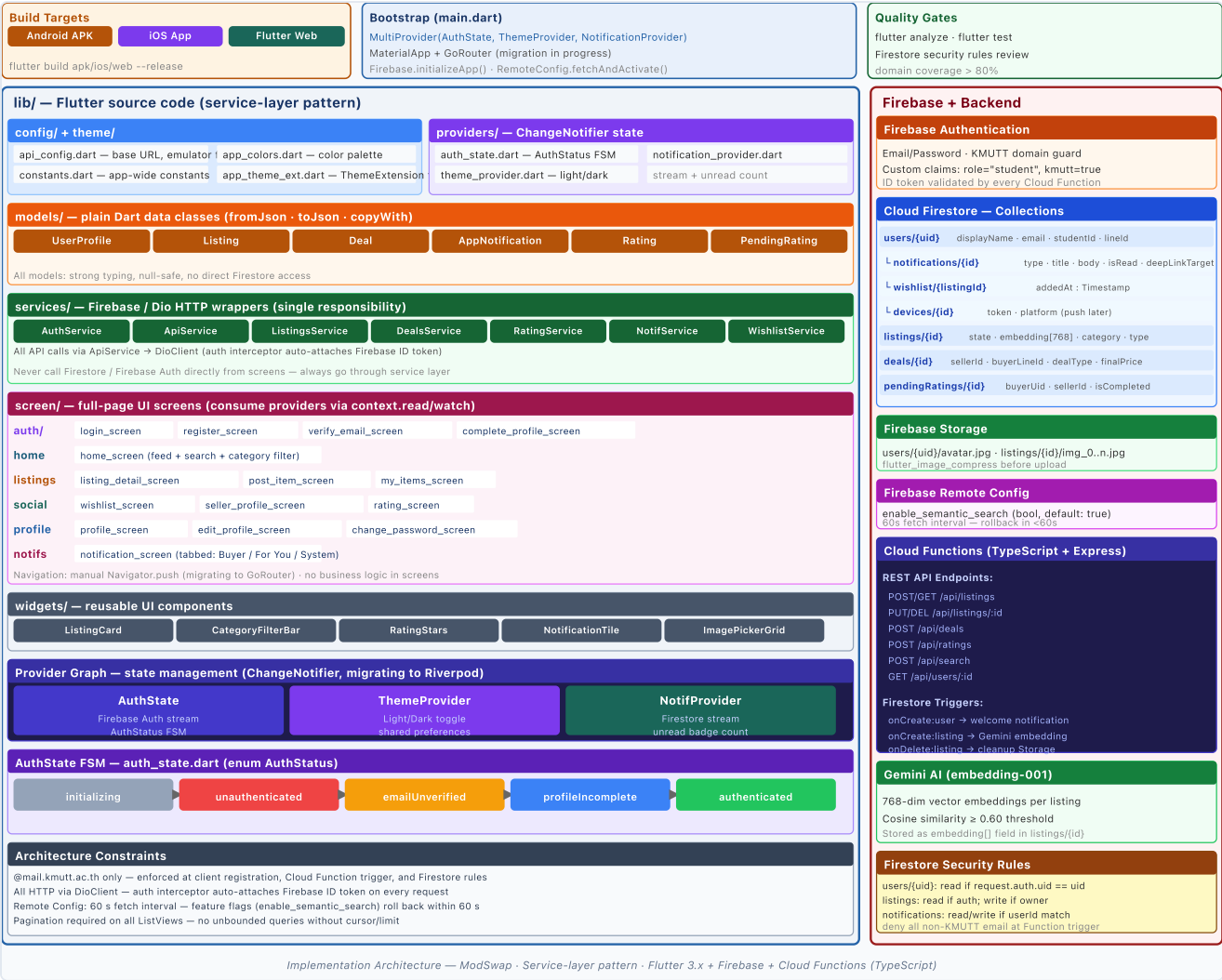
Layer Responsibilities

Layer	Responsibility
Presentation	Flutter widgets and screens; user interaction only; no business logic
Domain	Use cases (business rules); entities (data models); repository interfaces
Data	Concrete repository implementations; Firebase/API adapters
Backend	Server-side business logic; auth enforcement; AI search; Firestore triggers

Key Architectural Decisions

- **Separation of Concerns:** Each layer only depends on the layer directly beneath it
- **Dependency Inversion:** Domain layer defines interfaces; Data layer implements them
- **Feature-Based Organization:** Code organized by feature module (listings, deals, auth, etc.)
- **Server-Side Authority:** All sensitive operations (mark as sold, create deal) validated on Cloud Functions, never trusted from client alone

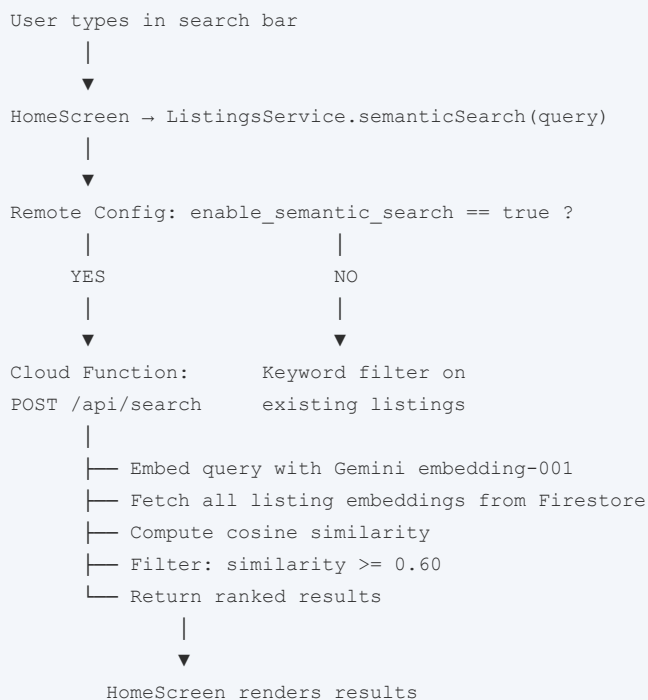
9. Implementation Architecture Diagram



Data Flow: Post a Listing



Data Flow: Semantic Search



State Management Flow



Document prepared by ModSwap Team · KMUTT · 2025–2026 Sorasit Laiget · Veerachai Sribuatim · Poowrin Trijakpranee · Chakkrapat Supanith · Siwakon Wanichsujit